A nilpotent-drift counterexample to the hypothesis-free L^1 Ornstein–Uhlenbeck spectrum extension

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Abstract

Kozhan proves that, when the adjoint drift generator A^* has an eigenvalue, the L^1 Ornstein–Uhlenbeck generator has spectrum the closed left half-plane and every point in the open left half-plane is an eigenvalue. He asks whether the theorem extends when $\sigma_p(A^*) = \varnothing$. The literal extension is false. Let S be the nilpotent left-shift semigroup on $E = L^2(0, 1)$ and take an injective Hilbert–Schmidt noise operator. Then A^* has no eigenvalues and the invariant Gaussian measure is nondegenerate. Nevertheless, after time one the Ornstein–Uhlenbeck semigroup is exactly the expectation projection. On the mean-zero subspace it is nilpotent, so its generator has entire resolvent; on constants its generator is zero. Thus the full L^1 generator has

$$\sigma(L) = \{0\},$$

not the closed left half-plane. An earlier paper records the nilpotent-shift semigroup identity in a BUC norm-continuity discussion; the spectrum inference and an explicit nondegenerate noise realization are supplied here.

1 Source question

The source is R. Kozhan, *L^1-spectrum of Banach space valued Ornstein–Uhlenbeck operators*, arXiv:1210.1287 [1]. Its main theorem and hypothesis are:

> $\overline{\mathbb{C}}_- = \{\lambda : \operatorname{Re}\lambda \le 0\}$ with each $\lambda \in \mathbb{C}_- = \{\lambda : \operatorname{Re}\lambda < 0\}$ being an eigenvalue. This result was extended to infinite-dimensional Banach spaces E in [12] under the assumption of eventual compactness of the semigroup $(S(t))_{t\ge 0}$, and in [2] under the assumption that the part of A^* in the reproducing kernel Hilbert space of μ_∞ has an eigenvalue $\gamma \in \mathbb{C}_-$. Theorem 1.1 of the present paper generalizes both of these results while requiring less effort to prove it. Also it can serve as an alternative simple coordinate-free proof of the corresponding finite-dimensional result of [9].
>
> **Theorem 1.1.** *If the point spectrum of A^* is not empty $\sigma_p(A^*) \neq \varnothing$, then the spectrum of the Ornstein–Uhlenbeck operator L coincides with $\overline{\mathbb{C}}_-$, and each $\lambda \in \mathbb{C}_-$ is its eigenvalue.*
>
> We remark that due to [11, Pr 2.5] we have $\sigma_p(A^*) \cap \{\lambda : \operatorname{Re}\lambda \ge 0\} = \varnothing$, and thus the condition $\sigma_p(A^*) \neq \varnothing$ of Theorem 1.1 is equivalent to the existence of an eigenvalue of A^*
>
> 2

The question appears at the top of the next page:

with negative real part. An extension of Theorem 1.1 for the case $\sigma_p(A^*) = \varnothing$ seems to be an open question so far.

Corollary 1.2. *Under the assumptions of Theorem 1.1, the Ornstein-Uhlenbeck semigroup $(P(t))_{t\geq 0}$ on $L^1(E, \mu_\infty)$ is norm discontinuous everywhere.*

The standing assumptions include a nondegenerate invariant centered Gaussian Radon measure μ_∞. We satisfy all of them. The conclusion fails in the strongest possible spectral way: the only spectral point is zero.

2 Proof intuition

The missing hypothesis permits drift semigroups whose generators have empty spectrum. The canonical example is the left shift on a finite interval. It forgets the entire initial state after one unit of time. Once the covariance has accumulated all of its mass, the associated Ornstein–Uhlenbeck transition therefore also forgets the initial state exactly, not merely asymptotically.

On $L^1(\mu_\infty)$, exact forgetting means that $P(t)$ is the expectation projection for $t \geq 1$. Removing constants leaves a genuinely nilpotent C_0-semigroup. The finite Laplace transform of a nilpotent semigroup is a resolvent at every complex number. This makes the spectral computation immediate once the Gaussian measure is verified to be nondegenerate.

3 The drift and the noise

Let $E = H = L^2(0, 1; \mathbb{R})$ and define

$$(S(t)f)(r) = \begin{cases} f(r+t), & r+t \leq 1, \\ 0, & r+t > 1. \end{cases}$$

This is a strongly continuous contraction semigroup and $S(t) = 0$ for $t \geq 1$.

Lemma 1. *If A is the generator of S, then $\sigma_p(A^*) = \varnothing$.*

Proof. On the complexification of E,

$$Af = f', \qquad D(A) = \{f \in H^1(0,1) : f(1) = 0\}.$$

Integration by parts gives

$$A^*g = -g', \qquad D(A^*) = \{g \in H^1(0,1) : g(0) = 0\}.$$

If $A^*g = \lambda g$, then $g(r) = ce^{-\lambda r}$. The boundary condition $g(0) = 0$ forces $c = 0$. This holds for every $\lambda \in \mathbb{C}$. $\square$

Choose an orthonormal basis $(e_n)_{n\geq 1}$ of E and define

$$Be_n = 2^{-n} e_n.$$

Then $B : H \to E$ is bounded, self-adjoint, injective, nonzero, and Hilbert–Schmidt. Put

$$Q_t = \int_0^t S(s)B^2S^*(s)\,ds, \qquad Q_\infty = \int_0^1 S(s)B^2S^*(s)\,ds. \tag{1}$$

Lemma 2. *Each Q_t is the covariance of a centered Gaussian Radon measure μ_t on E. The covariance Q_∞ is trace class with dense range, so its Gaussian measure μ_∞ is nondegenerate and invariant. Moreover, $Q_t = Q_\infty$ for $t \geq 1$.*

Proof. The operator in (1) is positive, self-adjoint, and

$$\operatorname{Tr} Q_t = \int_0^{\min(t,1)} \|S(s)B\|_{\mathrm{HS}}^2\,ds \leq \|B\|_{\mathrm{HS}}^2 < \infty.$$

Thus it is a Gaussian Radon covariance. For $x \neq 0$,

$$\langle Q_\infty x, x\rangle = \int_0^1 \|BS^*(s)x\|^2\,ds > 0.$$

Indeed, the continuous integrand is strictly positive at $s = 0$ because B is injective. Hence Q_∞ is injective; since it is self-adjoint, $\overline{\operatorname{Ran} Q_\infty} = E$. This is exactly nondegeneracy of the centered Gaussian measure.

The covariance identity

$$Q_\infty = Q_t + S(t)Q_\infty S^*(t)$$

follows by splitting the integral and using the semigroup property, and proves invariance. Finally, $S(s) = 0$ for $s \geq 1$, so (1) gives $Q_t = Q_\infty$ for every $t \geq 1$. □

4 Exact computation of the L^1 spectrum

Let $(P(t))_{t\geq 0}$ be the Ornstein–Uhlenbeck semigroup on $L^1(E, \mu_\infty)$ and let L be its generator.

Theorem 3. *For the drift and noise above,*

$$\sigma_p(A^*) = \varnothing, \qquad \sigma(L) = \{0\}.$$

In particular, the conclusion of source Theorem 1.1 does not extend to all cases with empty point spectrum of A^*.

Proof. The first assertion is Lemma 1. For $t \geq 1$ the Mehler formula, Lemma 2, and $S(t) = 0$ give

$$P(t)f(x) = \int_E f(y)\,d\mu_\infty(y) =: \Pi f. \tag{2}$$

This holds first for bounded Borel functions and hence for the unique contraction extension to $L^1(E, \mu_\infty)$.

Use the bounded invariant decomposition

$$L^1(E, \mu_\infty) = \mathbb{C}\mathbf{1} \oplus X_0, \qquad X_0 := \ker \Pi.$$

On constants $P(t)$ is the identity, so the generator is zero. The restricted semigroup $P_0(t) := P(t)|_{X_0}$ satisfies $P_0(t) = 0$ for $t \geq 1$ by (2). Let L_0 be its generator. For arbitrary $\lambda \in \mathbb{C}$ define the bounded operator

$$R_\lambda f := \int_0^1 e^{-\lambda t} P_0(t) f \, dt.$$

The standard semigroup differentiation calculation, whose endpoint term is $e^{-\lambda} P_0(1) = 0$, gives

$$(\lambda - L_0) R_\lambda f = f \quad (f \in X_0), \qquad R_\lambda (\lambda - L_0) g = g \quad (g \in D(L_0)).$$

Thus every $\lambda \in \mathbb{C}$ belongs to the resolvent set of L_0.

For $\lambda \neq 0$, the inverse of $\lambda - L$ on the displayed direct sum is

$$(\lambda - L)^{-1}(c\mathbf{1} + g) = \frac{c}{\lambda}\mathbf{1} + R_\lambda g.$$

Hence $\mathbb{C} \setminus \{0\} \subseteq \rho(L)$. On the other hand, $L\mathbf{1} = 0$, so zero is an eigenvalue. Therefore $\sigma(L) = \{0\}$. □

5 Prior overlap, limitations, and novelty audit

Van Neerven and Priola already used the same nilpotent shift in a discussion of norm discontinuity on BUC and observed $S(t) = 0$, $\mu_t = \mu_1$, and $P(t) = P(s)$ for $t, s \geq 1$ [2, Example 2.9]:

It should be observed that neither $S(t) \neq S(s)$ implies $\mu_t \perp \mu_s$ or conversely. An example of a periodic semigroup with period 1 such that $\mu_t \perp \mu_s$ for all $t, s \geqslant 1$ is given in [21, Example 3.8]. On the other hand, if $\dim E < \infty$, then for any choice of S and B the measures μ_t and μ_s are mutually absolutely continuous for all $t, s \geqslant t_0$.

We continue with two examples which show that $\|P(t) - P(s)\|_{\mathscr{L}(BUC^\circ(E))} < 2$ may occur for certain values of $t \neq s$.

Example 2.9 (Nilpotent S). Let $E = L^2(0,1)$ and let S be the nilpotent shift semigroup on $L^2(0,1)$, see for instance [12, page 120]. Then $S(t) = S(s) = 0$ and $\mu_t = \mu_s = \mu_1$ for all $t, s \geqslant 1$. Hence, $P(t) = P(s)$ for all $t \geqslant s \geqslant 1$.

Example 2.10 (Periodic S in finite dimensions). Let $H = E = \mathbb{R}^2$ and let S be the rotation group on $\mathbb{R}^2$. Let $B := I$, the identity operator on $\mathbb{R}^2$. Since $S^*(t) = S(-t)$ for all $t \geqslant 0$, the covariance operator of μ_t is given by $Q_t = tI$. Hence μ_t is the Gaussian measure on $\mathbb{R}^2$ with variance t. For $k = 0, 1, 2, \ldots$ and $f \in BUC^\circ(\mathbb{R}^2)$,

$$P^\circ(2k\pi)f(x) = \int f(S(2k\pi)x + y)\, d\mu_{2k\pi}(y) = \int f(x+y)\, d\mu_{2k\pi}(y).$$

That example is decisive supporting evidence rather than a later solution of the 2012 question: it predates the question and does not state the L^1 generator spectrum. The present packet makes the implication explicit, chooses an injective Hilbert–Schmidt noise so every standing Gaussian and nondegeneracy assumption is verified, and supplies the entire-resolvent calculation.

The counterexample exploits $\sigma(A) = \varnothing$, as is possible for a nilpotent C_0-semigroup generator. It does not settle the sharpened question in which one additionally assumes $\sigma(A) \neq \varnothing$ while retaining $\sigma_p(A^*) = \varnothing$.

A bounded search on August 13, 2026 checked all four lightweight run indexes, the exact title/id, the exact open-question phrase, and nilpotent-shift OU spectrum keywords. It found the source, standard finite-dimensional results, surveys, and the prior example above, but no paper stating this L^1 spectral counterexample. The result is recommended for expert review as a likely valid full negative answer to the question as literally worded, with deliberately limited novelty claims.

References

[1] R. Kozhan, *L^1-spectrum of Banach space valued Ornstein–Uhlenbeck operators*, Semigroup Forum 78 (2009), 547–553; arXiv:1210.1287.

[2] J. M. A. M. van Neerven and E. Priola, *Norm discontinuity and spectral properties of Ornstein–Uhlenbeck semigroups*, J. Evol. Equ. 5 (2005), 473–488.